

Solutions to Chapter 6 Exercises:

Exercise 1: Understanding RL Fundamentals

Supervised learning relies on labeled datasets with explicit input-output pairs, whereas reinforcement learning (RL) involves learning through interaction with an environment. RL does not require labeled data; instead, it uses rewards as feedback.

RL is particularly suitable for networking because:

- Network environments are dynamic and unpredictable
- Decisions have long-term consequences
- Feedback is often delayed (e.g., congestion effects)

Exercise 2: MDP Formulation

Example: Congestion Control

- **States (s):** Network conditions (RTT, packet loss, throughput)
- **Actions (a):** Increase, decrease, or maintain congestion window
- **Rewards (r):** Function of throughput and packet loss (e.g., high throughput – penalty for loss)
- **Policy (π):** Mapping from states to actions

This formulation allows the agent to learn optimal transmission strategies.

Exercise 3: Q-Learning Mechanics

The Q-learning update rule is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

- **Learning rate (α):** Controls how quickly the model updates its knowledge
- **Discount factor (γ):** Determines the importance of future rewards

A high γ prioritizes long-term rewards, while a low γ focuses on immediate gains.

Exercise 4: Exploration vs Exploitation

- **Exploration:** Trying new actions to discover better strategies
- **Exploitation:** Using known actions that yield high rewards

The ϵ -greedy strategy balances this by:

- Choosing a random action with probability ϵ
- Choosing the best-known action otherwise

Limitations:

- Fixed or poorly tuned ϵ may lead to suboptimal learning
- Random exploration may be inefficient in large state spaces

Exercise 5: RL in Networking Applications

Compared to AIMD:

- **Adaptability:** RL adapts to changing network conditions; AIMD follows fixed rules
- **Performance:** RL can achieve higher throughput by learning optimal policies
- **Risks:** RL may introduce instability or higher packet loss during training

RL offers better long-term optimization but requires careful design and training.